■ Black on White / White on Black Game

GRA 121 — Design Principles I

■ What You’re Doing

This game tests how your eyes see size, lightness, and contrast — not how you measure them. You’ll compare a black square on white and a white square on black to find out how value (light vs. dark) changes what we think we see. Even though both shapes start the same size, one will look bigger. Your job is to fix that optical trick.

■ Step-by-Step Instructions

1. Look Carefully
Two squares will appear side-by-side: one black square on a white background and one white square on a black background. They start exactly the same size — but they won’t look the same.

2. Adjust the Size
Use the slider or arrow buttons under the images to change the size of one of the squares. Go slowly and watch how your eyes react. Goal: Make both center squares look equal in size — not measured equal, but visually balanced.

3. Lock Your Answer
When you think they look the same, click Submit or Next. The game will record how close your choice is to the true optical balance.

4. Reflect for Credit
After finishing, write a short paragraph (3–5 sentences) answering: - Which square looked bigger or smaller at first? - What did you change to make them feel equal? - What does this tell you about how black and white affect perception in design? Example: “The white square looked smaller at first. I increased its size slightly until both felt equal. This showed me that light shapes seem to shrink against dark backgrounds.”

■ Why This Matters

Designers constantly deal with optical illusions — letters, icons, and layouts rarely look centered or balanced even when they are. This exercise trains your eye to trust perception, not just rulers.

■■ Game Info

- Attempts: You can play twice — the highest score counts.
- Time: About 5–10 minutes.

- Scoring: 10 pts for how close your adjustment is to the optical balance; 10 pts for your reflection paragraph.
- Accessibility: High-contrast, keyboard friendly, and screen-reader labeled.